Langsphere AI

Technical Architecture & Speech / Voice System Specification

Platform: React Native & Expo (v56) | Target Languages: Indic (Tamil, Hindi, Telugu, Malayalam, Kannada) & Global

1. Executive Summary & Core Platform Overview

Langsphere AI is an interactive educational platform designed to teach children Indic languages (Tamil, Hindi, Telugu, Malayalam, Kannada) and international languages. The application integrates dynamic Generative AI lesson creation, interactive tracing canvases, gamified XP progression, real-time voice recording/pronunciation analysis, and parent monitoring capabilities.

2. Complete Technical Stack Matrix

Category	Technologies & Libraries	Version	Role & Key Responsibilities
Frontend Framework	React Native, Expo	0.85.3 / 56.0.3	Cross-platform runtime engine for Android, iOS, and Web targets.
Router & Navigation	expo-router	56.2.5	File-based routing with tab bar, stack navigation, and route guards.
UI & Styling	NativeWind, Tailwind CSS	4.2.4 / 3.3.2	Utility-first styling system for consistent component design tokens.
Animations & 3D	Reanimated, Three.js, R3F	4.3.1 / 0.184.0	60 FPS micro-animations, particle effects, and 3D mascot rendering.
Generative AI Engine	@google/generative-ai	0.24.1	Powers *gemini-flash-lite-latest* for dynamic placement tests, AI stories, and chatbot mascot.
Backend & Auth	Firebase Auth & Firestore, Express	12.13.0 / 5.2.1	User login, streak tracking, parent dashboard, and custom Node API endpoints.
Voice Output (TTS)	expo-speech, Web SpeechSynthesizer, Google TTS	56.0.3	Multi-tier text-to-speech output with audio stream fallback.
Voice Input (STT)	Web Speech API (webkitSpeechRecognition)	Native Web	Real-time microphone capture, Indic Unicode regex filtering, and pronunciation scoring.

3. Deep Dive: Voice & Speech System Architecture

The voice ecosystem in Langsphere AI is built around two key pipelines: **Text-to-Speech (TTS)** for natural audio narration and **Speech-to-Text (STT)** for user voice recording and pronunciation evaluation.

A. Text-to-Speech (TTS) — Audio Output Engine

Location: `utils/speech.ts`
The system maps internal language identifiers to standard BCP-47 codes:

- **Tamil:** `ta-IN` | **Hindi:** `hi-IN` | **Telugu:** `te-IN`
- **Malayalam:** `ml-IN` | **Kannada:** `kn-IN` | **English:** `en-US`

3-Tier Execution Strategy:

1. **Tier 1 (Mobile Native):** Invokes `Speech.speak(text, { language: langCode, rate: 0.85 })` using Expo Speech. Speed is calibrated to 0.85 for language learners.
2. **Tier 2 (Browser Native):** On Web platforms, checks `window.speechSynthesis` for an installed native voice matching the language prefix (e.g. `ta` or `hi`).
3. **Tier 3 (Cloud Fallback):** If native engines lack voice packs for regional Indic scripts, constructs an audio stream URL and plays it via Web Audio API (`new window.Audio(url).play()`):

```
https://translate.google.com/translate_tts?ie=UTF-8&q;={TEXT}&tl;={LANG}&client;=tw-ob
```

B. Speech-to-Text (STT) & Pronunciation Evaluation

Location: `components/practice/VoiceRecordingUI.tsx` & `app/game-pronounce.tsx`
Voice input processing relies on real-time browser microphone streaming combined with Indic Unicode character filtering:

1. **Microphone Stream Capture:** Instantiates `window.webkitSpeechRecognition` configured with continuous listening and interim result streaming.
2. **Indic Unicode Regex Sanitization:** To process complex scripts accurately, non-alphanumeric noise is removed while retaining specific South Asian script ranges:

```
// Unicode script ranges: Tamil (\u0B80-\u0BFF), Hindi (\u0900-\u097F), Telugu (\u0C00-\u0C7F) const
cleanSpoken = currentTranscript.toLowerCase().replace(
/[^\w\s\u0B80-\u0BFF\u0900-\u097F\u0C00-\u0C7F\u0D00-\u0D7F\u0C80-\u0CFF]/g, ' ' );
```

3. **Accuracy Match Scoring:** Spoken word tokens are checked against target phrase tokens. Matches trigger live UI highlighting and compute an overall accuracy percentage.

4. Feature & Module Integration Matrix

Module / Screen	Primary Source Path	Voice & Speech Integration Mechanism
Pronunciation Game	app/game-pronounce.tsx	STT microphone stream, token matching, accuracy score rendering.
Vocabulary Flashcards	components/practice/VoiceRecordingUI.tsx	TTS audio demonstration + interactive STT pronunciation test.
AI Mascot Chatbot	utils/ai.ts	Gemini AI text answers rendered into audible mascot speech via TTS.
Reading Comprehension	utils/stories.ts	AI story generator with automated paragraph-by-paragraph TTS narration.